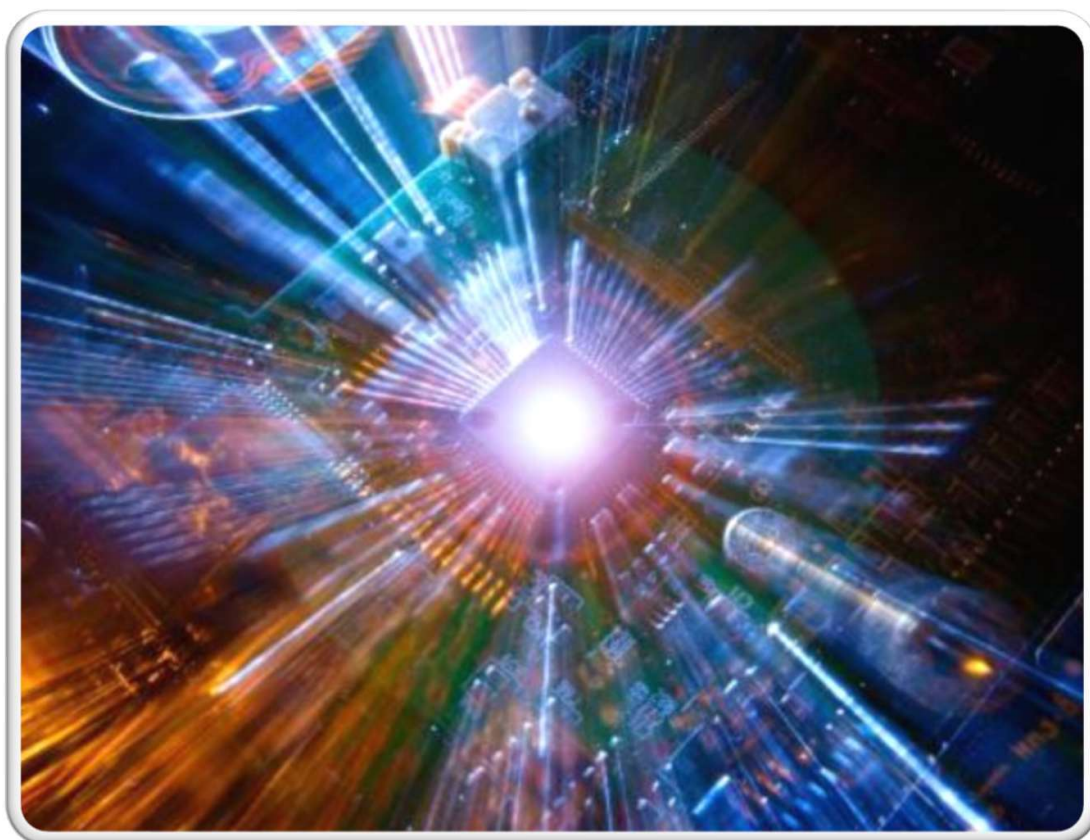




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Innovations in Physics Exploring Advances in Materials, Optics, and Technology



Editors

Dr. Jyoti Rajput

Dr. Mukesh Kumar

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ISBN:978-81-975304-2-5

2024

Brief Profiles of Editors



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CHAPTER-3

Materials for 3D Printed Radiation Dosimeters

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Abstract. Radiation therapy is the cornerstone of cancer treatment in modern medicine. To ensure accurate radiation dose delivery, measurement of the radiation dose is vital. Radiation dosimetry is performed in several ways for ensuring safe and effective dose to the tumor. The radiation dosimeters along with the phantoms are being used for estimation of radiation absorbed dose. Phantoms are generally made up of materials replicating the radiological properties of human tissues. With customization in radiation therapy, the dosimetry systems need custom designed phantoms and dosimeters. 3D printing technology plays a role in such situations. The technology enables customized 3D objects of the desired materials for radiation dosimetry. Several materials such as PLA, PETG, ABS, biodegradable material, nanocomposites etc. have been studied in different radiation dosimetry and related applications. These materials have been found to be dosimetrically suitable, possessing radiological properties nearly equivalent to human tissues. The 3D printed materials have been utilized to model the dosimetry phantoms, dosimeter supports and the patient support systems. The 3D printing technology in radiation therapy is currently not so popular due to multiple factors such as cost, expertise and availability of materials. There is a need for making 3D printing affordable and more user friendly. The integration of these devices with smart material and artificial intelligence (AI) has the potential to revolutionize the radiation dosimetry in clinics, industry and environment.

The present chapter discusses the role of 3D printing in radiation therapy and radiation dosimetry along with various kinds of materials used for 3D printed radiation dosimeters, phantoms and support systems. The chapter also presents an insight onto the current status and future indications for 3D printing in the radiation dosimetry.

Keywords: 3D printing, radiation dosimetry, radiotherapy, phantoms, dosimeters.

Introduction

Cancer is one of the widest spreading diseases of the current times [1]. Cancer is a group of diseases characterized by unwanted growth and spread of abnormal cells in the body. It is a major health concern in modern times as it is related to lifestyles [2], [3]. Cancer may occur in any part of the body and is generally characterized by its histology and primary site of its occurrence. Cancer treatment is a multimodal approach mainly involving surgery, chemotherapy and radiation therapy. Immunotherapy, hormonal therapy, target therapy, stem cells implant, artificial intelligence and quantum nanodots etc. are some of the developing techniques for customized

cancer treatment [4], [5], [6], [7]. Surgery aims to remove the tumor and surrounding subclinical disease. It is employed mainly for solid tumors. Chemotherapy uses cytotoxic drugs to kill the cancer cells and stop its spread. Radiation therapy makes use of the high energy ionizing radiations to kill the cancer cells [8]. All these modalities are combined either in pairs or all together depending upon several factors like tumor type, stage, age, general conditions etc. [9].

Radiation therapy is one of the major modalities for cancer treatment. It uses a wide spectrum of ionizing radiation ranging from electromagnetic to particle radiation and low energy (MeV) to high energy (GeV) radiation [10]. The radiation therapy works on a universal goal of delivering maximum dose to the tumor i.e. to increase the tumor control probability (TCP) while minimizing the dose to surrounding normal tissues i.e. to keep the normal tissue complication probability (NTCP) as low as reasonably practicable [11]. Radiation therapy has shown great results in tumor control with small normal tissue morbidities. The treatments comprising small fields such as stereotactic radiosurgery/radio therapy (SRS/RT) have been found to have nearly equivalent clinical controls to surgery in some brain tumors. Radiotherapy has proven to be a highly effective non-invasive technique of cancer care [12].

The radiation beams however doesn't have selective approach and cause the cell kill in each cell types. It is important to accurately localize the tumors and its vicinity and place it accurately under the radiation beam for accuracy of the treatment and higher TCP [13], [14]. This goal is achieved by conventional marking and positioning methods integrated with modern imaging techniques like on board x-ray cone beam computed tomography (CBCT) and other on board imaging modalities [15], [16]. Along with this it is important to measure the radiation beams parameters accurately. The radiation absorbed dose along the beam central axis, beam off axis ratios, beam flatness and symmetry, patient specific quality assurance must be measured accurately to ensure the accuracy of radiation dose. Here comes the role of radiation dosimetry, the process of measuring radiation dose [17].

Radiation dosimetry plays a vital role in ensuring the accuracy of radiation therapy. Various types of dosimeters have been employed in the dosimetry of therapeutic radiation beams ranging from the air vented and gas filled ionization chambers, the thermoluminescent dosimeters, optically stimulated luminescent dosimeters, solid state dosimeters, chemical dosimeters being some of the classical examples [17], [18]. Some of the modern dosimeters include the Cherenkov detectors, optical fiber detectors, wireless dosimeters, plastic scintillators, 3D gel dosimeters and 3D printed dosimeters etc. These dosimeters have applications in small stereotactic, ultra-high dose rate field dosimetry of x-ray and particle beam radiations [19].

Along with the dosimeters, the role of phantoms is equally vital in radiation therapy quality assurance (QA) and dosimetry. The phantoms are considered as replicates of human anatomy and are used for experimental and measurement purposes [20]. With evolution in the treatment technology and techniques there has been a surge in the phantom designs. Various commercial vendors have proposed phantoms capable of customized QA of the system and handling wide variety of radiation dosimeters for radiation measurements. Phantoms are generally made up of plastics and polymers having radiological properties nearly equivalent to human soft tissues or water e.g. Polymethylmethacrylate (PMMA), polyester, polyvinyl chloride (PVC), polyvinyl alcohol (PVA) and solid water etc. [21].

With customization in radiation dosimetry, the importance of customization in dosimetry has raised. Patients in advanced radiation therapy may undergo specific procedures and therapeutic sessions. In such cases the generalized dosimetry systems may have limited role and more customized and specific dosimeters and phantoms may be required. The fabrication of such phantoms and dosimeters may not be commercially viable and needs local fabrication. Three-dimensional (3D) printing) is one of the most suitable options for such fabrications. With 3D printing the customization can be incorporated with accuracy as required in radiation therapy [22].

3D printing or additive manufacturing has now become widespread. Its use in medicine has become common. Innovations in 3D printing technology have significantly enhanced the field of radiation therapy, particularly in the development of custom-fit devices and radiation dosimetry tools [23], [24], [25], [26]. It has been used to design new phantoms, tissue compensators, bolus and dosimeters. The 3D printers have shown promising results in radiation therapy applications. As mentioned earlier, the phantoms and dosimeters for radiation therapy require materials with specific radiological properties. The proceeding sections shall include 3D printing technology, materials required for 3D printing the radiation dosimetry and their applications. The chapter will also discuss the future of 3D printing in radiation dosimetry.

The technology of 3D Printing

3d printing is the process of sequentially layering the materials for creation of a 3d object using the digital information. The technology has acquired a vital space in the industry, manufacturing, healthcare, aerospace and many more. The 3d printing involves several steps. The 3D design of the object to be created is prepared

using digital programs such as computer aided design (CAD). The CAD designs are then converted into printable formats such as stereolithography or OBJ.

The printers are loaded with the required materials and printing parameters such as layer height, temperature, printing speed etc. are set. The printers then print in layer-by-layer fashion. Multiple materials may be employed if a layer is designed such. There are various printing techniques such as fused deposition modelling (FDM), stereolithography (SLA), selective layer sintering (SLS), digital light processing (DLP), multi jet fusion (MJF), binder jetting and metal printing [27], [28].

Materials or 3D printed dosimeters

The selection of materials for 3D printed radiation dosimeters is critical to ensure accurate measurement and evaluation of radiation doses. These materials must possess specific properties to mimic human tissue response to radiation and provide reliable dosimetric data. The materials should be physically stable under irradiation conditions. Here are some commonly used materials and innovative options being explored for 3D printed radiation dosimeters:

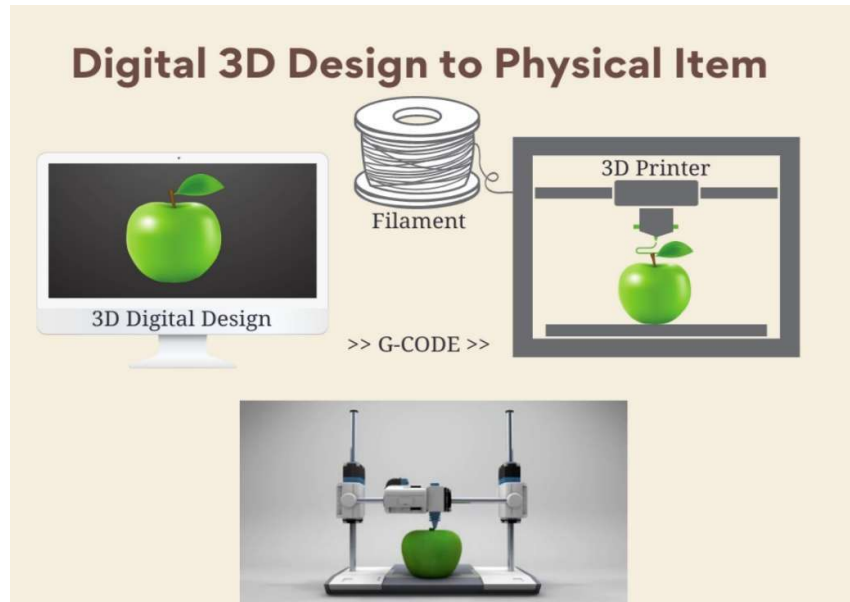


Fig. 1. The 3D printing process showing various components of the process in making of the real 3D model. (from : <https://northerngeometry.com>)

Common Materials for 3D Printed Dosimeters

1. Polylactic Acid (PLA)

PLA is a polymer thermoplastic derived from organic sources such as corn starch and sugar cane. It can also be produced from petrochemical plastics. PLA is one of the most produced bioplastics. PLA, due to its thermoplastic and bioplastic properties and near tissue equivalence, has several applications in radiation therapy. It has been used to cast customized dosimetry phantoms and dosimetry components. It has also been used to prepare the support structures such as dosimeter holders and immobilization devices for customized radiation therapy [28], [29], [30], [31].

- **Advantages:** PLA possesses properties such as Biodegradable, easy to print, low cost, dimensional accuracy and accessibility.
- **Applications:** PLA has been used for creating phantoms, supporting structures and simple dosimeter prototypes.
- **Limitations:** PLA itself is not radiation sensitive hence can't be used directly as a radiation dosimeter. PLA is strong and brittle hence may not be suitable for high stress conditions. Moreover, its thermal resistance is low and hence can't be used in high temperature conditions.

2. Acrylonitrile Butadiene Styrene (ABS)

ABS composed of acrylonitrile, butadiene and styrene is an amorphous terpolymer. The materials are processed by injection molding, thermos forming and extruding methods. ABS is a common thermoplastic material which is often used in manufacturing light and rigid products. It has been used to prepare the automotive parts, wheel covers, enclosures and head protectors. ABS has been one of the earliest materials to be used for 3D printing. It is a popular material for 3D printing known for its durability, thermal strength and strength. The physical and thermal properties of this polymer can be modulated by modifying the concentrations of the constituents [32], [33], [34].

- **Advantages:** ABS is durable, heat-resistant (high melting point 220-250), and easy to print, flexible, customizability and surface finish compound. Its strength and durability make it ideal for making phantoms and support material for radiation therapy dosimetry.
- **Applications:** ABS has been found to be suitable for more robust dosimeter housings and components.
- **Limitations:** ABS emits fumes when heated, requiring proper ventilation during printing. Its radiation sensitivity is like PLA hence can't be used directly as a radiation dosimeter.



Fig. 2. Formlab 4B 3D printer. (from: <https://formlabs.com>)

3. Polyethylene Terephthalate Glycol (PETG)

PETG is a thermoplastic polymer possessing properties such as significant chemical resistance, formability for manufacturing and durability. The polyethylene terephthalate (PET) was initially invented for textiles but, due to its too soft nature, the PET was mixed with glycol to produce PETG. It has applications in high temperature industries due to its heat-resistant properties. PETG has been used for packaging and bottle forming. Due to its adhesion properties, minimal deformation, chemical resistance and odor lessness, PETG has become a suitable material for 3D printed dosimetry phantom and supports [35], [36], [37], [38].

- **Advantages:** PETG is robust with high impact and chemical resistance, ease of printing and translucency. It is flexible and has ease of printing.
- **Applications:** PETG has been used for construction of phantoms, dosimeter cases and support structures.
- **Limitations:** PETG can be challenging to print due to warping and requires precise temperature control. It has a higher cost and requires fine tuning during printing to avoid its oozing and straining.

4. Radiation-Sensitive Gels and Resins

The specially designed radiation-sensitive gels and resins are specially designed to behave as radiation dosimeters. The materials possess radiosensitivity and result in some measurable changes in irradiation. The radiation induced changes may be in physical or chemical state like change in color, opacity or emission of light etc. Commonly employed radiation sensitive gels and resins are radiochromic films, BANG gel, polyvinyl alcohol glutaraldehyde (PVA-GTA) incorporating radiochromic dyes. Some polymer gels, radiochromic resins, hydrogels and silicon-based materials also come under this category [39], [40], [41], [42], [43], [44], [45]. These materials can directly be used as radiation dosimeters and incorporated in a substrate to act as a dosimeter. All these materials possess the possibility of getting engineered using the 3D printing to obtain desired sizes, shapes and concentration of the radiosensitive component. Such customization allows the use of these 3D printed dosimeters and dosimetry phantoms in patient specific applications of dosimetry such as small and very small fields, particle therapy and high dose rate radiation [31].

- **Advantages:** These materials Can be engineered to change properties (e.g., color, opacity) in response to radiation exposure.
- **Applications:** The radiation sensitive gels and resins have been used to create highly sensitive and precise dosimeters that provide visual or measurable feedback on radiation exposure in specialized and advanced radiation therapy.
- **Examples:** Radiochromic gels and resins that darken upon exposure to radiation.
- **Limitations:** The radiation sensitive gels and resins can be printed effortlessly; however, the cost of the materials and impact of environmental conditions may affect the fabrication process.

5. Tissue-Equivalent Materials

The materials are specially fabricated to mimic human tissues, be it soft tissues like muscles and adipose tissues or the bones. These materials possess radiological equivalence. The materials are highly crucial in radiotherapy dosimetry for accurate dose measurements especially in customized radiation therapy. The polyurethane (PU), hydrogel composites, agar and gelatin composites and some silicon resins possess tissue equivalence properties. These materials can be custom printed to mimic the shape and size of the human tissues under study [46], [47], [48], [49], [50], [51]. Hence, these materials post customized 3D printing possess the properties of an ideal dosimetry phantoms capable of exactly predicting the patient dose.

- **Properties:** These materials are designed to mimic the radiological properties of human tissues, ensuring accurate dose measurements.
- **Applications:** These are essential for creating realistic phantoms and dosimeters that replicate human tissue response.
- **Examples:** Hydrogel composites, silicone-based materials, and specialized polymers.
- **Limitations:** These materials possess limited mechanical strength, have high cost and precise formulation is required to produce the exact compound.

Table 1. The comparative analysis of different materials used for 3D printing in radiation dosimetry.

Property/Feature	PLA	ABS	PETG	Radiation-Sensitive Gels and Resins	Tissue-Equivalent Materials
Ease of Printing	High	Moderate	High	Moderate	Moderate
Cost	Low	Moderate	Moderate	High	High
Mechanical Strength	Moderate	High	High	Varies	Moderate to High
Flexibility	Low	Moderate	Moderate to High	Low to Moderate	Moderate
Thermal Resistance	Low	High	Moderate	Low to Moderate	Moderate
Chemical Resistance	Low	Moderate	High	Varies	Moderate to High
Radiation Sensitivity	Low	Low	Low	High	Moderate to High

Biodegradability	Yes	No	No	No	No
Common Applications	Phantoms, Prototypes, Support Structures	Phantoms, Housings, Fixtures	Phantoms, Housings, Fixtures	Dosimeters	Phantoms, Dosimeters
Environmental Impact	Low (Biodegradable)	Moderate	Moderate	High	Moderate
Print Settings Complexity	Low	High	Moderate	High (varies with material composition)	Moderate to High
Post-Processing	Easy (minimal)	Moderate (sanding, acetone smoothing)	Easy (minimal)	Moderate (depends on gel/resin properties)	Moderate (coatings, treatments)
Customizability	High	High	High	High	High
Notable Limitations	Brittle, Low Radiation Resistance	Emits Fumes, Warping Issues	Higher Cost, Stringing Issues	Expensive, Handling Complexity	Cost, Specialized Handling
Future Potential	Moderate (Enhanced Composites)	High (Smart Coatings, Functional Additives)	High (Smart Technologies, composites)	Very High (Innovative Dosimetric Solutions)	High (Advanced Tissue Simulation)

Composite and Hybrid Materials

Metal-Polymer Composites

Metal-polymer composites are advanced materials that combine the properties of metals and polymers to create a hybrid material with enhanced characteristics. These composites are gaining attention in the field of radiation dosimetry due to their improved mechanical properties, radiation sensitivity, and customization capabilities through 3D printing. The polymer materials such as PLA, ABS, PETG and nylon with metals such as Copper (Cu), Calcium (Ca), Aluminum (Al) etc. can be used in 3D printing to produce enhanced materials with metals having higher radiation sensitivity and polymer providing the core structure [52], [53], [54], [55]. The Ca doped materials have been utilized to cast the tissue equivalent phantoms that can simulate the bones and soft tissues more accurately [56]. The 3D printed nanoparticle-polymer composites have also been tested as shielding materials [57].

- **Advantages:** The metal polymer composites combine the structural benefits of polymers with the radiation absorption properties of metals providing a custom radiation dosimeter.
- **Applications:** Metal polymer composites have been used for creating dosimeters with enhanced sensitivity and durability. Further, these materials can be used to incorporate nanocomposites and integrated with sensors and wireless for real time wireless dosimetry applications.
- **Examples:** Composites with embedded metallic nanoparticles (e.g., silver, gold) to improve radiation interaction.
- **Limitations:** These materials possess high cost; the presence of metals may affect the flow of printing and adhesion properties of the composite.

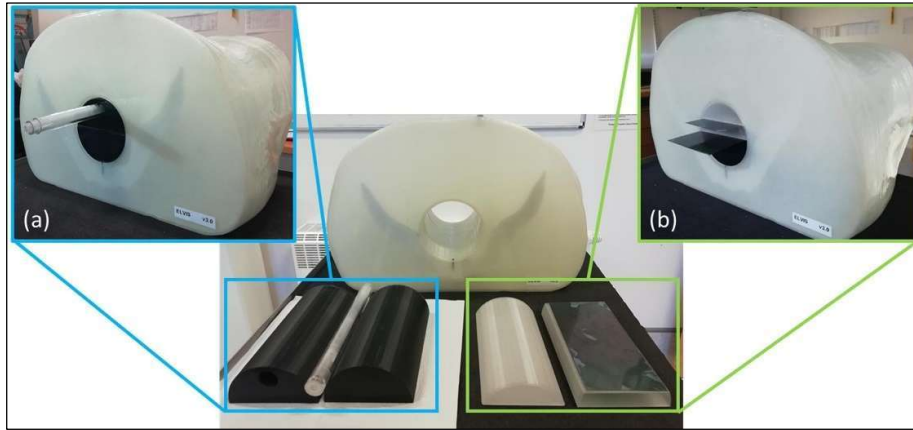


Fig. 3. The 3D printed pelvis phantom for customised accurate dosimetry of the pelvis region. The custom designed inserts for ionisation chamber (a) and garfhrmonic films (b) (from: [58]).

Carbon-Based Materials

Carbon-based materials are increasingly being explored for use in 3D printed radiation dosimeters due to their unique properties, such as high strength, electrical conductivity, thermal stability, and radiation resistance. This article discusses various carbon-based materials, their properties, advantages, limitations, and applications in radiation dosimetry. Common carbon-based materials like graphene, carbon nanotubes, carbon fiber, carbon black and graphite have shown applications in dosimetry [58], [59], [60], [61]. The 3D printing of dosimeters and dosimetry components based on carbon has been found to be able to produce customized, highly sensitive, thermally and mechanically stable with high electrical conductivity and resistance to radiation damage dosimetry products [62].

- **Advantages:** The carbon-based materials possess good radiation response along with their rigid mechanical properties. These materials are light weight, radiation damage resistant and thermally stable.
- **Applications:** The carbon-based materials have been utilized in advanced dosimeters that require high durability and precise measurement capabilities.
- **Examples:** Graphene-infused polymers and carbon nanotube composites.
- **Limitations:** The materials have a high cost; these may affect the printing process and have complex handling and processing processes and need specialized equipment.

Innovations and Future Directions

Smart Materials

Smart material includes Piezoelectric Materials, Thermochromic Materials, Photochromic Materials, Shape-Memory Materials, Electroactive Polymers, and Magnetorheological and Electrorheological Fluids that responds dynamically to external effects such as radiations. These materials have the ability to change their properties or behavior in response to environmental changes, making them highly valuable for advanced applications, including radiation dosimetry [63], [64].

- **Advantages:** The smart materials can integrate sensors and wireless communication capabilities to provide real-time feedback on radiation exposure. These possess high dynamic response, and enhanced radiation sensitivity. The materials have versatile physico-chemical properties so that these can be tailored to specific dosimetric needs and integrated with other technologies.
- **Applications:** These materials are used in next-generation dosimeters for continuous monitoring and data transmission.
- **Examples:** Materials with embedded RFID chips or wireless sensors.
- **Limitations:** The materials are yet to be generalized for 3D printed radiation dosimeters and hence there is complexity in its manufacturing. These materials are limited by high cost and limited durability.

Nano-Engineered Materials

Nano-engineered materials, which incorporate nanotechnology to manipulate materials at the molecular or atomic level, offer significant potential in the field of 3D printed radiation dosimeters [65]. These materials exhibit unique properties such as enhanced strength, increased surface area, improved radiation sensitivity, and superior electrical and thermal conductivity. Nanoparticles, nanocomposites and quantum dots etc. have applications as 3D printed nano-engineered materials.

- **Advantages:** These materials utilize nanotechnology to enhance the sensitivity and accuracy of dosimeters.
- **Applications:** The nano composite materials are suitable for highly sensitive applications and precise dosimetry in complex environments.
- **Examples:** Nanoparticle-infused gels and resins, nanocomposite materials.
- **Limitations:** Nanocomposites are limited by cost of production and integration of nano materials. Their manufacturing, handling and safety further limits the application of these materials.

Biodegradable Materials

Biodegradable materials are becoming increasingly important in various fields due to their environmental benefits. In radiation dosimetry, biodegradable materials offer a sustainable alternative for creating dosimeters that can break down naturally after their useful life. This article explores various biodegradable materials used in 3D printed radiation dosimeters, their properties, advantages, limitations, and applications. Biodegradable materials such as PLS, polycaprolactone (PCL), polyglycolic acid (PGA) and starch and cellulose based materials have been tested for their applications in 3D printed radiation dosimeters.

- **Advantages:** These materials are environmentally friendly and capable of breaking down naturally after use and hence pose no issues of environmental safety.
- **Applications:** These materials have been used in temporary dosimeters for short-term monitoring, reducing environmental impact.
- **Examples:** Biodegradable polymers and composites.
- **Limitations:** These materials have limited thermal stability, lower mechanical strength and higher degradation rate.

Future of 3d printing in radiation therapy

3D printing has an enormous future in the field of radiation therapy. The currently employed materials have tissue equivalence, good mechanical and thermal properties which makes them suitable for printing phantoms, dosimeters and support systems. The future of 3D printing has significant advancements poised to enhance the precision, personalization, and efficiency of cancer treatments.

The incorporation of AI and networking in the 3D printed devices shall revolutionize the dosimetry making it wireless, online, predictable and controllable. Application of such devices shall ease the difficulties of real time in-vivo dosimetry. In such cases the determination of patient dose in real time shall be more accurate ensuring the accurate treatment. The applications of 3D printed systems in ultra-high dose rate therapies and particle therapies shall necessarily pave the way for its better clinical implementation and enhanced accuracy.

Conclusion

The development of 3D printed radiation dosimeters involves a careful selection of materials that can accurately measure radiation doses while providing the necessary mechanical and chemical properties. Common materials like PLA, ABS, and PETG offer a good starting point, but advanced options such as radiation-sensitive gels, tissue-equivalent materials, and innovative composites are paving the way for more precise and reliable dosimetry. As research and technology progress, the future of 3D printed dosimeters looks promising, with smart and nano-engineered materials leading the charge towards more sophisticated and effective radiation measurement tools.

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